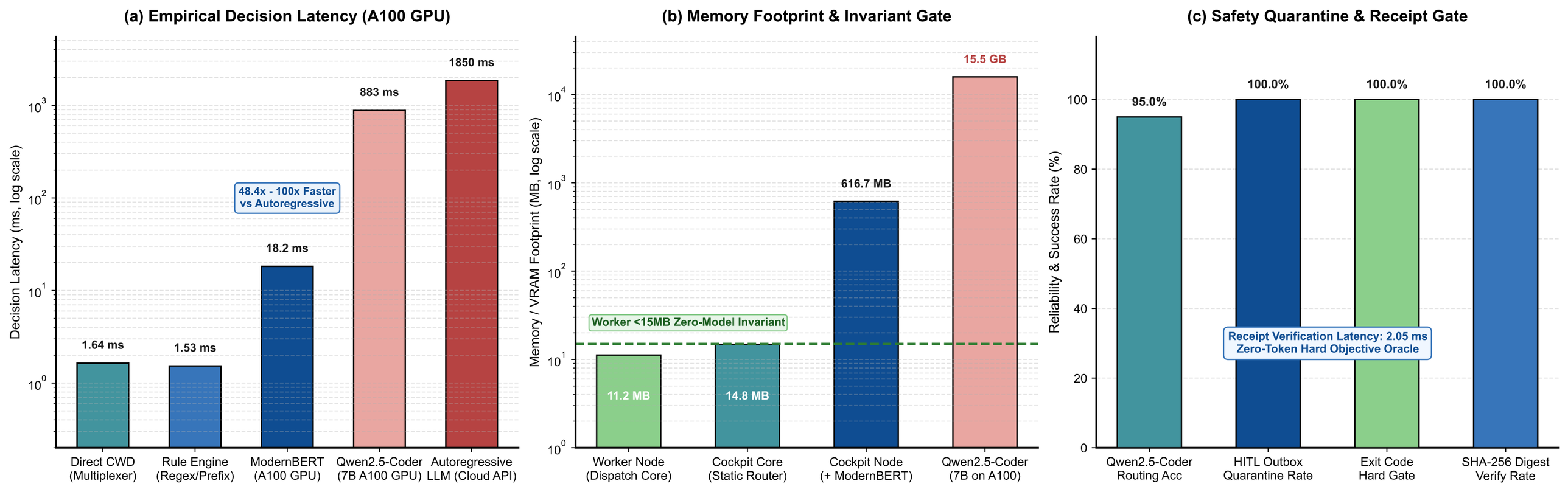
(a) Empirical Decision Latency (A100 GPU)
Decision Latency (ms, log scale)
1.64 ms
1.53 ms
18.2 ms
883 ms
1850 ms
48.4x - 100x Faster
vs Autoregressive
Direct CWD
(Multiplexer)
Rule Engine
(Regex/Prefix)
ModernBERT
(A100 GPU)
Qwen2.5-Coder
(7B A100 GPU)
Autoregressive
LLM (Cloud API)
(b) Memory Footprint & Invariant Gate
Memory / VRAM Footprint (MB, log scale)
Worker <15MB Zero-Model Invariant
11.2 MB
14.8 MB
616.7 MB
15.5 GB
Worker Node
(Dispatch Core)
Cockpit Core
(Static Router)
Cockpit Node
(+ ModernBERT)
Qwen2.5-Coder
(7B on A100)
(c) Safety Quarantine & Receipt Gate
Reliability & Success Rate (%)
95.0%
100.0%
100.0%
100.0%
Receipt Verification Latency: 2.05 ms
Zero-Token Hard Objective Oracle
Qwen2.5-Coder
Routing Acc
HITL Outbox
Quarantine Rate
Exit Code
Hard Gate
SHA-256 Digest
Verify Rate